Improved Integral Attack on ChiLow-32 Exploiting the Inverse of the ChiChi Function*

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Abstract. The protection of executable code in embedded systems requires efficient mechanisms that ensure confidentiality and integrity. Belkheyar *et al.* recently proposed the Authenticated Code Encryption (ACE) framework, with `ChiLow`-($32 + \tau$) as the first instantiation of ACE2 at EUROCRYPT 2025. The design of `ChiLow`-($32 + \tau$) is based on a 32-bit tweakable block cipher with a quadratic nonlinear layer, known as ChiChi (denoted by $\chi\chi$), and a nested tweak key schedule optimized for secure code execution under strict query limits.

In this work, we study the resistance of `ChiLow` to integral cryptanalysis. We identify new integral distinguishers in both the single-tweak and related-tweak models. Using these results and a nested strategy to recover all round tweaks, we present a key-recovery attack on 7 out of 8 rounds of `ChiLow`. The central contribution of our work is that it resolves the challenge of deriving the master key from the recovered round tweaks, an open problem highlighted by the designers and in a recent cryptanalysis by Peng *et al.* The attack on 7 rounds requires $2^{6.32}$ chosen ciphertexts, has a time complexity of about $2^{121.75}$ encryptions, and requires negligible memory.

Keywords: Cryptanalysis · Integral attack · Authenticated Code Encryption · ChiLow · ChiChi

1 Introduction

The security of embedded systems has become a major concern as such devices are deployed in sensitive tasks ranging from consumer electronics to industrial control and automotive systems. In many cases, the protection of executable code is crucial. Adversaries who can read or tamper with program instructions may steal intellectual property, inject malicious behavior, or bypass security checks. To address this threat, designers have proposed code encryption as a mechanism to ensure that only authorized code is executed on the device. Beyond confidentiality, integrity must also be guaranteed, so that modified instructions can be detected before execution.

Authenticated Code Encryption (ACE) was introduced by Belkheyar *et al.* [BDG+25] as a dedicated framework to meet these requirements. The ACE structures combine decryption of short instruction words with the computation of authentication tags. This approach enables secure and efficient code execution in resource constrained environments.

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Three main variants were proposed, called ACE-1, ACE-2, and ACE-3, which represent different trade offs between performance and security margin.

Among them, ACE-2 aims at achieving low latency while preserving a sufficient security margin. The primitive `ChiLow`-$(32 + \tau)$ is the first instantiation of ACE-2. It is built from a 32-bit tweakable block cipher and a related tweakable function for tag generation, and it follows a nested tweak key schedule. The design is optimized for embedded processors that must handle instruction words of 32 bits, while meeting strict query limits: at most 2^{40} in total and 2^8 per tweak. These limits model practical conditions for secure boot and firmware update in constrained devices.

Because of these features, the study of `ChiLow`-$(32 + \tau)$ is of particular interest. On the one hand, it addresses a real world problem that arises in the secure execution of code on lightweight platforms. On the other hand, its design choices, such as the low degree non linear layer $\chi\chi_n$, raise natural questions about its resistance to advanced cryptanalytic techniques. In particular, the designers have not provided a comprehensive security analysis against integral attacks [DKR97, KW02], which are among the most powerful tools for evaluating ciphers with low degree round functions [Lai94]. This leaves an important gap in the evaluation of `ChiLow`-$(32 + \tau)$.

Integral cryptanalysis is especially interesting for `ChiLow` because of the quadratic nature of its non linear layer, which suggests that strong integral properties may propagate over several rounds. At the same time, the strict query limits imposed by the designers make the attack challenging, since extending distinguishers backward in the single-tweak model may exceed the allowed data complexity. This combination makes integral cryptanalysis both a natural and a non trivial choice for assessing the security of `ChiLow`.

Contributions. In this work we investigate the resistance of `ChiLow`-$(32 + \tau)$ against integral cryptanalysis. Our main contributions are:

- We identify several low-data complexity integral distinguishers for `ChiLow` in both the single-tweak and related-tweak models. We also experimentally verified the validity of all of our single-tweak integral distinguishers. Table 1a briefly describes our distinguishers.
- We present a key recovery attack on 7 rounds of `ChiLow` by combining these distinguishers. The attack requires $2^{6.32}$ chosen ciphertexts, runs in time dominated by $2^{121.75}$ encryptions, and needs negligible memory.
- According to the designers, recovering the later round tweaks does not easily compromise the master key or reveal round tweaks for different tweak values, even if an attacker manages to obtain them. We have overcome this challenge by implementing a nested approach. In this method, we initially use a longer distinguisher to recover the tweaks of the last rounds. Then shorter distinguisherss will be exploited to recover the tweaks of the early rounds as well as some bits of the master key. In this way, we can retrieve the master key for the recovered round tweaks.

Organization. The rest of the paper is organized as follows. Section 2 recalls notation, the tools for integral analysis, and the structure of `ChiLow`. Section 3 presents our integral distinguishers in both the single-tweak and related-tweak models. **??** describes the key recovery attacks and analyzes complexities. Finally, Section 5 concludes the paper.

2 Background

In this section, we provide a brief overview of the concepts and tools that we use for analyzing `ChiLow`. We review the core idea of integral attack, the specification of `ChiLow`-$(32 + \tau)$, and summarize the notations we use throughout the paper.

Table 1: Summary of results on ChiLow-(32+τ).

(a) Integral distinguishers.

Model	#R	#Dist.	#Active Bits.	Time and Data complexity
Single-tweak	2	1	2	2^2
Single-tweak	3	9	4	2^4
Single-tweak	3	12	3	2^3
Single-tweak	3.5	17	8	2^8
Related-tweak	6	1	8+25	2^{33} total, 2^8/tweak

(b) Key-recovery attack. In this table, "CC" is short for chosen ciphertext, "KP" for known plaintext, "ST" for single-tweak, "MT" for multi-tweak, "Cond" for conditional, "Bord" for borderline, and "MITM" for meet-in-the-middle.

#R	Time	Data	Memory	Setting	Method	Source
6	$\mathbf{2^{98.1}}$	$\mathbf{2^{5.6}}$ **CC**	**negligible**	**ST**	**Integral**	**Subsection 4.1**
6	$\mathbf{2^{89.78}}$	$\mathbf{2^{6.32}}$ **CC**	**negligible**	**ST**	**Integral**	**Subsection 4.2**
6	2^{120}	2^{40} CC	0	MT	Cond.cube	[PHH+25]
6	2^{34}	$2^{33.58}$ CC	$2^{54.28}$	MT	Bord.cube	[PHH+25]
7	$2^{127.75}$	2^8 CC	2^{23}	ST	Integral	[PHH+25]
7	$\mathbf{2^{121.75}}$	$\mathbf{2^{6.32}}$ **CC**	**negligible**	**ST**	**Integral**	**Subsection 4.2**
8	$2^{125.37}$	82 KP	$2^{98.36}$	ST	MITM	[LNV25]
8	$2^{122.6}$	97 KP	NA	ST	MITM	[GWZ25]
8	$2^{120.35}$	162 KP	$2^{98.34}$	ST	MITM	[LNV25]
8	$2^{111.42*}$	165 KP	$2^{101.37}$	ST	MITM	[LNV25]
8	$2^{108.6}$	196 KP	NA	ST	MITM	[GWZ25]

*The master key will be recovered with probability $2^{-3.75}$.

2.1 Integral Attack

Integral cryptanalysis studies how sums of outputs behave when the inputs vary over a structured set. In the bit oriented setting we choose a set of inputs that covers all 2^d assignments of a selected group of d input bits, while the remaining bits stay constant. If an output bit sums to zero over this set, we say it is balanced or it has a zero-sum property. We denote an r round integral property by $A \xrightarrow{r\text{-round}} B$, where A is the index set of active input bits and B is the index set of balanced output bits as in our notation section.

To search such properties at scale we use the division property. We follow the conventional and bit based division properties of Todo and Morii [Tod15, TM16] together with the automated method proposed by Xiang et al. [XZBL16]. The division property tracks whether every monomial that contains a given set of variables can appear in the algebraic normal form of an intermediate bit. If no such monomial can appear, then the corresponding output bit is balanced when we sum over all values of the active inputs.

We encode the propagation of the division property through each round as a mixed integer linear programming (MILP) model. Each round is decomposed into its elementary operations, such as XOR, AND, and copy. We describe each operation locally by linear constraints from [XZBL16] and combine them to represent the whole round. Chaining these constraints across several rounds yields a global feasibility model for the chosen number of rounds. If the model shows that no monomial involving the active inputs and the key can reach a target output bit, then that bit is guaranteed to be balanced for any fixed key.

Unlike the common approach in block cipher cryptanalysis, where the focus is typically on the encryption algorithm, the design strategy of `ChiLow` directs the analysis toward the decryption algorithm. Therefore, our attack targets and investigates the decryption process.

We use two data models. In the single-tweak model the tweak is fixed and only the ciphertext bits are active at the input. In the related-tweak model both the ciphertext and selected tweak bits may be active. We denote such a property by $A, T \xrightarrow{r\text{-round}} B$, where T is the set of active tweak indices. The data cost equals $2^{|A|}$ in the single-tweak model and $2^{|A|} \cdot 2^{|T|}$ in the related-tweak model, subject to the limits given by the design.

Our search procedure is as follows. We fix a round number and a candidate active set A at the input. We build the MILP instance that captures the division property through all components of those rounds, setting the objective to minimize the sum of output bits. After solving the instance, if the objective value equals one (corresponding to a unit vector in the output), the associated bit is not necessarily balanced. We then iteratively update the model by excluding all previously identified unit vectors and continue solving until no further unit vectors are found. For the given active set A, the output bits not associated with any recovered unit vector constitute the complete set. B. Repeating this process over all small active sets yields the catalog of distinguishers we report in this paper. Due to the low data complexity of our distinguishers, we finally verify some of the distinguisher by direct experiments, which confirms that the predicted output positions are indeed balanced for the required number of queries.

2.2 Notation

Table 2 provides a summary of the notations used throughout this work.

Table 2: Notation summary.

Notation	Meaning
$X[i]$	i^{th} bit of n-bit word X (with $X[0]$ as LSB)
$\overline{X[i]}$	$X[i] \oplus 1$
$A \xrightarrow{r\text{-round}} B$	r-round single-tweak integral distinguisher,
	A: active input bits, B: balanced output bits
$A, T \xrightarrow{r\text{-round}} B$	r-round related-tweak integral distinguisher,
	T: active tweak bits
$\mathcal{E}_r(P, T_0^{(6)}, \ldots, T_0^{(6-r+1)})$	r-round decryption of P under given round tweaks

2.3 Specification of `ChiLow`-(32+τ)

`ChiLow`-(32+τ) instantiates the ACE2 (Authenticated Code Encryption) construction, defined in [BDG+25], and consists of two closely related tweakable primitives built on the same core structure: a 32-bit tweakable block cipher D_{32} and a tweakable function D'_{32}. D_{32} corresponds to the decryption algorithm to decrypt an encrypted instruction (ciphertext) C of 32 bits. Besides C, it takes as input the 128-bit master key $K = K_0 \,\|\, K_1 \,\|\, K_2 \,\|\, K_3$ and a 64-bit tweak $T = T_0 \,\|\, T_1$, and outputs the 32-bit structure (plaintext) P. D'_{32} truncated to τ bits ($\tau \in \{8, 16\}$) computes an authentication tag for an encrypted instruction of 32 bits. It takes the same inputs exactly as D_{32}. Figure 1 shows an overall view of the full `ChiLow`-($32 + \tau$) decryption and tag computation.

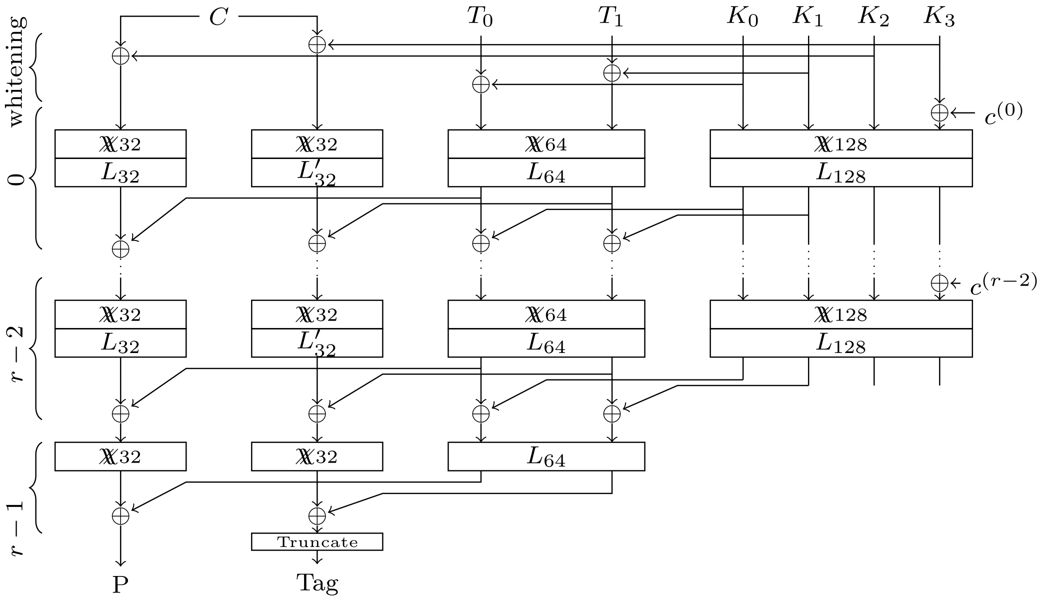

Figure 1: A full `ChiLow`-$(32+\tau)$ decryption [BDG+25]

`ChiLow` uses a nested tweak-key schedule architecture. The internal state is divided into three distinct parts that are updated in parallel over rnd rounds (claimed secure with $\mathsf{rnd} = 8$):

1. **Key State Update:** It is initialized with the master key K, which is updated as follows.
$$K^{(i+1)} = L_{128}(\ \chi_{128}(K^{(i)} \oplus c^{(i)})) \tag{1}$$
where $c^{(i)}$ is a 32-bit round constant added to bits 96-127 of K, and L_{128} and χ_{128} are linear and nonlinear transforms, respectively.

2. **Tweak State Update:** It is initialized with $T \oplus (K_0 \,\|\, K_1)$, and is updated by the following round function incorporating material from the key state[1]:
$$T^{(i+1)} = L_{64}(\ \chi_{64}(T^{(i)} \oplus (K_0 \,\|\, K_1)^{(i+1)})\ ) \tag{2}$$
The last round is composed only of the linear mapping, i.e.,
$$T^{(\mathsf{rnd})} = L_{64}(T^{(\mathsf{rnd}-1)}) \tag{3}$$

3. **Cipher States Update:** it involves two independent 32-bit states, X for D_{32}, initialized with $C \oplus K_2$, and X' for D'_{32}, initialized with $X \oplus K_3$. Both are updated by round functions $\mathcal{R}[T^{(i+1)}](\cdot)$ and $\mathcal{R}'[T^{(i+1)}](\cdot)$, respectively, which incorporate material from the tweak state, as follows:
$$\begin{aligned} D_{32} : X^{(i+1)} &= \mathcal{R}[T^{(i+1)}](X^{(i)}) = L_{32}(\chi_{32}(X^{(i)})\ ) \oplus T_0^{(i+1)} \\ D'_{32} : X'^{(i+1)} &= \mathcal{R}'[T^{(i+1)}](X'^{(i)}) = L'_{32}(\chi_{32}(X'^{(i)})\ ) \oplus T_1^{(i+1)} \end{aligned} \tag{4}$$
For the last round, only the non-linear mapping is applied as follows:
$$\begin{aligned} X^{(\mathsf{rnd})} &= \chi_{32}(X^{(\mathsf{rnd}-1)}) \oplus T_0^{(\mathsf{rnd})} \\ X'^{(\mathsf{rnd})} &= \chi_{32}(X'^{(\mathsf{rnd}-1)}) \oplus T_1^{(\mathsf{rnd})} \end{aligned} \tag{5}$$

[1] The cipher specification of `ChiLow`-$(32+\tau)$ slightly differs from the figure and test vectors given in [BDG+25]. This formula is consistent with the figure and the test vectors of the original paper.

Table 3: Offsets for the linear layers.

Linear Map	α	β_0	β_1	β_2
L_{32}	11	5	9	12
L'_{32}	11	1	26	30
L_{64}	3	1	26	50
L_{128}	17	7	11	14

Non-Linear Layer $\chi\chi_n$. The nonlinear transformation is a bijective function designed for even state sizes $n = 2m$, based on the well-known χ functions [Dae95]. It is defined as follows:

$$y_i = \begin{cases} x_i + \overline{x}_{i+1}x_{i+2} & i < m-3 \text{ or } m < i < n-2 \\ x_m + \overline{x}_{m-2}x_0 & i = m-3 \\ x_{m-1} + \overline{x}_0 x_1 & i = m-2 \\ \overline{x}_{m-3} + \overline{x}_m \overline{x}_{m+1} & i = m-1 \\ x_{m-2} + \overline{x}_{m+1}x_{m+2} & i = m \\ x_{n-2} + \overline{x}_{n-1}x_{m-1} & i = n-2 \\ x_{n-1} + \overline{x}_{m-1}x_m & i = n-1 \end{cases} \tag{6}$$

Linear Layer L_n. All linear layers in `ChiLow`-(32+τ) satisfy the relation given below, with the corresponding parameters specified for each case in Table 3.

$$y_i = x_{\alpha i+\beta_0} + x_{\alpha i+\beta_1} + x_{\alpha i+\beta_2} \tag{7}$$

Security Claims. The designers claim $\pm\widetilde{\text{prp}}$ security of the underlying block cipher D_{32} and prf security of the truncated version D'_{32} to τ bits, provided that the total number of queries satisfies $q \leq 2^{40}$ and the number of queries per tweak satisfies $q_t \leq 2^8$ [BDG+25].

2.3.1 Properties of the Non-linear Layer

The non-linear transformation is a bijective function designed for even state sizes $n = 2m$, based on the widely used χ functions [Dae95, BDPA13, DEMS21, ABD+24]. Since $\chi\chi_n$ is bijective for even $n = 2m$, the inverse $\chi\chi_n^{-1}$ exists for all inputs. Unlike the forward map, which has quadratic algebraic degree, the inverse has a much higher degree. It is conjectured that the algebraic degree of each non-trivial component of $\chi\chi_n^{-1}$ is m [BDG+25]. Moreover, each coordinate x_i of $X = \chi\chi_n^{-1}(Y)$ depends non-linearly on almost all coordinates of Y, which means that the inverse diffusion is very strong. There are, however, four special cases where the dependency is only linear:

- x_{m-3} depends linearly on y_{m-1},
- x_{m-2} depends linearly on y_m,
- x_{m-1} depends linearly on y_{m-2},
- x_m depends linearly on y_{m-3}.

These properties highlight both the appeal and the difficulty of mounting an integral attack on this design. The low algebraic degree of the forward nonlinear layer supports the existence of integral distinguishers over several rounds. By contrast, the high algebraic degree of the inverse tends to destroy simple parity relations when propagated backward. In the current analysis, we do not expand $\chi\chi_{32}^{-1}$ symbolically. Instead, we treat it as a black box and evaluate it using the reference implementation, alternating with L_{32}^{-1}. A more effective approach is to exploit the few coordinates of χ_{32}^{-1} that remain linear and to perform parity checks only on output bits where balance is expected to persist. This strategy keeps the required data within the limits set by the design and reduces the risk of false positives when searching over tweak candidates.

3 Integral Distinguishers for ChiLow-(32+τ)

Here, we discuss the details of our distinguishers on ChiLow-(32 + τ).

3.1 Integral Distinguishers in the single-tweak single-key model

To find integral distinguishers for ChiLow-(32+τ), we utilize the Mixed-Integer Linear Programming (MILP) modeling framework for the conventional division property [Tod15, TM16] described in [XZBL16]. We have exhaustively searched for all possible 3-round integral distinguishers involving four active bits and identified 9 such distinguishers, with 32 balanced output bits. Each of the following input sets independently yields a 3-round integral distinguisher, where the entire 32-bit output state is balanced:

$$\{5, 7, 9, 11\},\ \{6, 8, 10, 12\},\ \{7, 9, 11, 14\},\ \{8, 10, 12, 14\},\ \{14, 17, 19, 21\},$$
$$\{21, 23, 25, 27\},\ \{22, 24, 26, 28\},\ \{23, 25, 27, 29\},\ \{23, 25, 27, 30\}.$$

Similarly, we conducted a comprehensive search for all possible 3-round integral distinguishers involving three active bits. The maximum number of balanced bits observed is 5, which occurs in two cases. Additionally, we identified 10 distinguishers that result in 3 balanced bits. These findings are listed below.

$$\begin{aligned}
\{21, 23, 25\} &\xrightarrow{\text{3-round}} \{2, 3, 14, 25, 26\}, & \{27, 29, 31\} &\xrightarrow{\text{3-round}} \{5, 14, 22, 25, 28\},\\
\{3, 5, 7\} &\xrightarrow{\text{3-round}} \{0, 4, 24\}, & \{6, 8, 10\} &\xrightarrow{\text{3-round}} \{5, 24, 27\},\\
\{7, 9, 11\} &\xrightarrow{\text{3-round}} \{0, 9, 12\}, & \{9, 11, 13\} &\xrightarrow{\text{3-round}} \{7, 11, 31\},\\
\{12, 14, 17\} &\xrightarrow{\text{3-round}} \{6, 21, 24\}, & \{17, 19, 21\} &\xrightarrow{\text{3-round}} \{14, 19, 23\},\\
\{18, 20, 22\} &\xrightarrow{\text{3-round}} \{6, 18, 27\}, & \{19, 21, 25\} &\xrightarrow{\text{3-round}} \{2, 14, 26\},\\
\{22, 24, 26\} &\xrightarrow{\text{3-round}} \{21, 25, 30\}, & \{26, 28, 30\} &\xrightarrow{\text{3-round}} \{1, 10, 22\}.
\end{aligned}$$

Since certain 3-round integral distinguishers exist with 4 active bits and all 32 output bits balanced, the possibility of a longer distinguisher was not unrealistic. Therefore, we conducted an exhaustive search for a 3.5-round distinguisher—i.e., a 3-round distinguisher extended by one nonlinear layer—with 8 non-adjacent active bits. The results are as follows:

$$\begin{aligned}
\{1, 3, 5, 9, 13, 15, 17, 30\} &\xrightarrow{\text{3.5-round}} \{25, 26\}, & \{4, 6, 8, 10, 12, 14, 17, 19\} &\xrightarrow{\text{3.5-round}} \{22, 23\},\\
\{10, 14, 17, 19, 21, 23, 25, 29\} &\xrightarrow{\text{3.5-round}} \{1, 5\}, & \{14, 17, 19, 21, 23, 25, 27, 29\} &\xrightarrow{\text{3.5-round}} \{1, 2\},\\
\{14, 16, 20, 22, 24, 26, 28, 30\} &\xrightarrow{\text{3.5-round}} \{23, 24\}, & \{14, 16, 20, 22, 24, 26, 28, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\},\\
\{14, 16, 20, 22, 24, 26, 29, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\}, & \{14, 16, 20, 22, 24, 27, 29, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\},\\
\{14, 16, 20, 22, 25, 27, 29, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\}, & \{14, 16, 20, 23, 25, 27, 29, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\},\\
\{14, 16, 21, 23, 25, 27, 29, 31\} &\xrightarrow{\text{3.5-round}} \{23, 24\}, & \{1, 3, 5, 7, 9, 11, 13, 15\} &\xrightarrow{\text{3.5-round}} \{25\},\\
\{1, 3, 5, 7, 9, 11, 13, 31\} &\xrightarrow{\text{3.5-round}} \{25\}, & \{2, 4, 6, 8, 10, 12, 29, 31\} &\xrightarrow{\text{3.5-round}} \{25\},\\
\{10, 14, 17, 19, 21, 23, 25, 27\} &\xrightarrow{\text{3.5-round}} \{1\}, & \{10, 14, 17, 19, 21, 23, 25, 28\} &\xrightarrow{\text{3.5-round}} \{1\}.
\end{aligned}$$

The choice of non-adjacent active bits is motivated by the χ function, where the non-linear term predominantly appears as $x_i x_{i+1}$. Based on this property, we focused our search for 3.5-round distinguishers within a more promising subspace. Nonetheless, as our random search indicates, distinguishers with adjacent active bits may also exist—for example, $\{0, 5, 7, 9, 11, 13, 16, 17\} \xrightarrow{\text{3.5-round}} \{25\}$.

3.2 Integral Distinguishers in the related-tweak single-key model

In the related-tweak single-key model, distinguishers can span over a greater number of rounds, due to the more relaxed constraints on the number of queries. For instance, an integral distinguisher for 6 rounds of `ChiLow`-(32+τ) is discovered, involving 8 active bits in the ciphertext and 25 active bits in the tweak. This conforms to the data limitations, specifically a total of 2^{40} queirs with no more than 2^8 queries per tweak. The distinguisher is described as follows:

$$\begin{cases} \text{ciphertext: } \{8, 10, 12, 14, 21, 23, 25, 27\} \\ \quad\;\;\text{tweak: } \{0, 2, 8, 10, 12, 14, 21, 23, 25, 27, 34, 36, 38 \xrightarrow{\text{6-round}} \{0, \ldots, 31\} \\ \qquad\qquad 40, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62\} \end{cases}$$

4 Key Recovery Attack on `ChiLow`-(32+τ)

In this section, first we introduce a primary 6-round integral key recovery attack. Then, we propose an improved version for the 6-round attack and its extension to a 7-round key recovery attack.

4.1 Primary 6-round Attack

In this attack, we attach three rounds of key recovery to the three-round distinguishers (see Figure 2). Unlike conventional integral attacks, which rely only on the longest distinguishers, we also use shorter ones, such as two-round distinguishers, to prune the number of guesses and reduce the overall complexity. In particular, the two-round distinguishers help in recovering the earlier round tweaks.

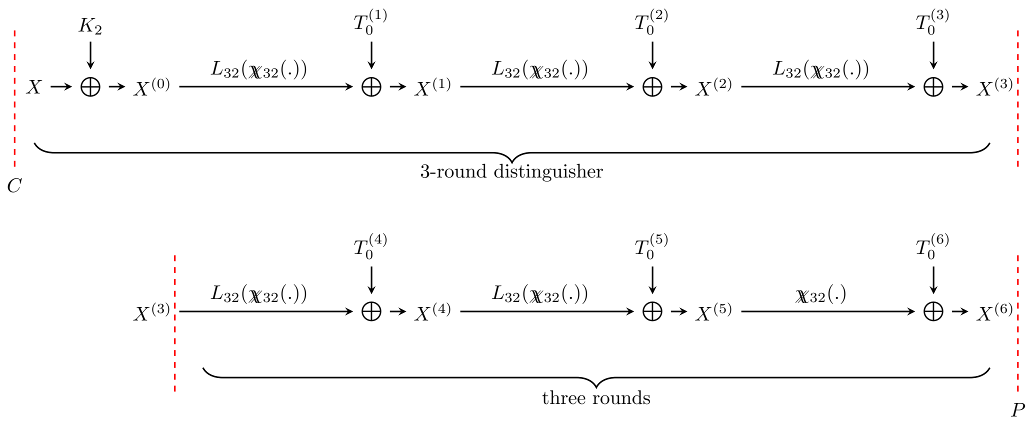

Figure 2: Attack on 6-round `ChiLow`-(32+τ).

Our key recovery proceeds by propagating backward across several rounds from the plaintext side. This requires repeated evaluation of L_{32}^{-1} interleaved with χ_{32}^{-1}. The algorithm 1 and algorithm 2 present the full procedure of our key recovery attack and provide a clear basis for the subsequent complexity analysis. We define the following functions used in algorithm 1, and algorithm 2:

$$\mathcal{E}_2(P, T_0^{(6)}, T_0^{(5)}) := \chi_{32}^{-1}\Big(L_{32}^{-1}\big(\chi_{32}^{-1}(P \oplus T_0^{(6)}) \oplus T_0^{(5)}\big)\Big), \tag{8}$$

$$\mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}) := \chi_{32}^{-1}\Big(L_{32}^{-1}\big(\mathcal{E}_2(P, T_0^{(6)}, T_0^{(5)}) \oplus T_0^{(4)}\big)\Big), \tag{9}$$

$$\mathcal{E}_4(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}, T_0^{(3)}) := \chi_{32}^{-1}\Big(L_{32}^{-1}\big(\mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}) \oplus T_0^{(3)}\big)\Big), \tag{10}$$

$$\mathcal{E}_5(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}, T_0^{(3)}, T_0^{(2)}) := \chi_{32}^{-1}\Big(L_{32}^{-1}\big(\mathcal{E}_4(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}, T_0^{(3)}) \oplus T_0^{(2)}\big)\Big). \tag{11}$$

Moreover, we define

$$\mathcal{D}_i(C, T_0^{(i)}, ..., T_0^{(1)}, K_2) := L_{32}\Big(\chi_{32}\big(\mathcal{D}_{i-1}(C, T_0^{(i-1)}, ..., T_0^{(1)}, K_2)\big)\Big) \oplus T_0^{(i)}, \tag{12}$$

for $i \in \{2, 3, 4, 5\}$; and

$$\mathcal{D}_1(C, T_0^{(1)}, K_2) := L_{32}\big(\chi_{32}(C \oplus K_2)\big) \oplus T_0^{(1)}, \tag{13}$$

$$\mathcal{D}(C, T_0^{(6)}, ..., T_0^{(1)}, K_2) := \chi_{32}\big(\mathcal{D}_5(C, T_0^{(5)}, ..., T_0^{(1)}, K_2)\big) \oplus T_0^{(6)}. \tag{14}$$

Figure 3 illustrates the distinct phases of our 6-round attack, which serves as the core of the subsequent 7-round attack. In what follows, we describe the attack in detail. The attack comprises four phases.

First phase: By employing 3-round integral distinguishers and extending them three rounds forward, we determine candidate values for the tweak states $T_0^{(4)}$, $T_0^{(5)}$ and $T_0^{(6)}$. The distinguishers used in this phase are:

$$\{21, 23, 25\} \xrightarrow{\text{3-round}} \{2, 3, 14, 25, 26\}, \tag{15}$$

$$\{21, 23, 25, 27\} \xrightarrow{\text{3-round}} \{0, \dots, 31\}. \tag{16}$$

Note that the active bits in the first distinguisher form a subset of those in the second. This relationship is beneficial for reducing the number of required queries. Hereafter, we refer to the distinguisher in Equation 15 as *Distinguisher I*, and the one in Equation 16 as *Distinguisher II*.

First, we choose three distinct sets of 2^4 ciphertexts which are active in bit numbers $\{21, 23, 25, 27\}$ and constant at other positions; and query for their corresponding plaintexts. Then, for 2^{96} possible values for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$, we compute the parity of the third-round output for a subset of the first input set according to Distinguisher I; and we continue with the values leading to the balanced property in bit numbers 2, 3, 14, 25, and 26. We expect $2^{91} (= 2^{96} \times 2^{-5})$ values for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$ pass this filter, with time complexity $2^{98} (= 2^{96} \times 2^3 \times \frac{3}{6})$ encryptions.

Afterward, for 2^{91} filtered values for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$, we compute the parity across the entire first input set. Exploiting the expected 32-bit balanced property according to Distinguisher II, this reduces the number of possible values for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$ to $2^{59} (= 2^{91} \times 2^{-32})$. The corresponding time complexity is $2^{93} (= 2^{91} \times 2^3 \times \frac{3}{6})$. For the candidate values for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$, we also calculate the parity of the second and third input sets and reduce the number of candidates to one with time complexity $2^{62} (\approx ((2^{59} \times 2^4) + (2^{27} \times 2^4)) \times \frac{3}{6})$ encryptions, due to the 2×32-bit condition. Hence, we get one candidate for $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$ on average, where the time complexity is dominated by $2^{98} + 2^{93} + 2^{62}$ encryptions and we estimate the total time complexity of this phase as $2^{98.1}$ encryptions.

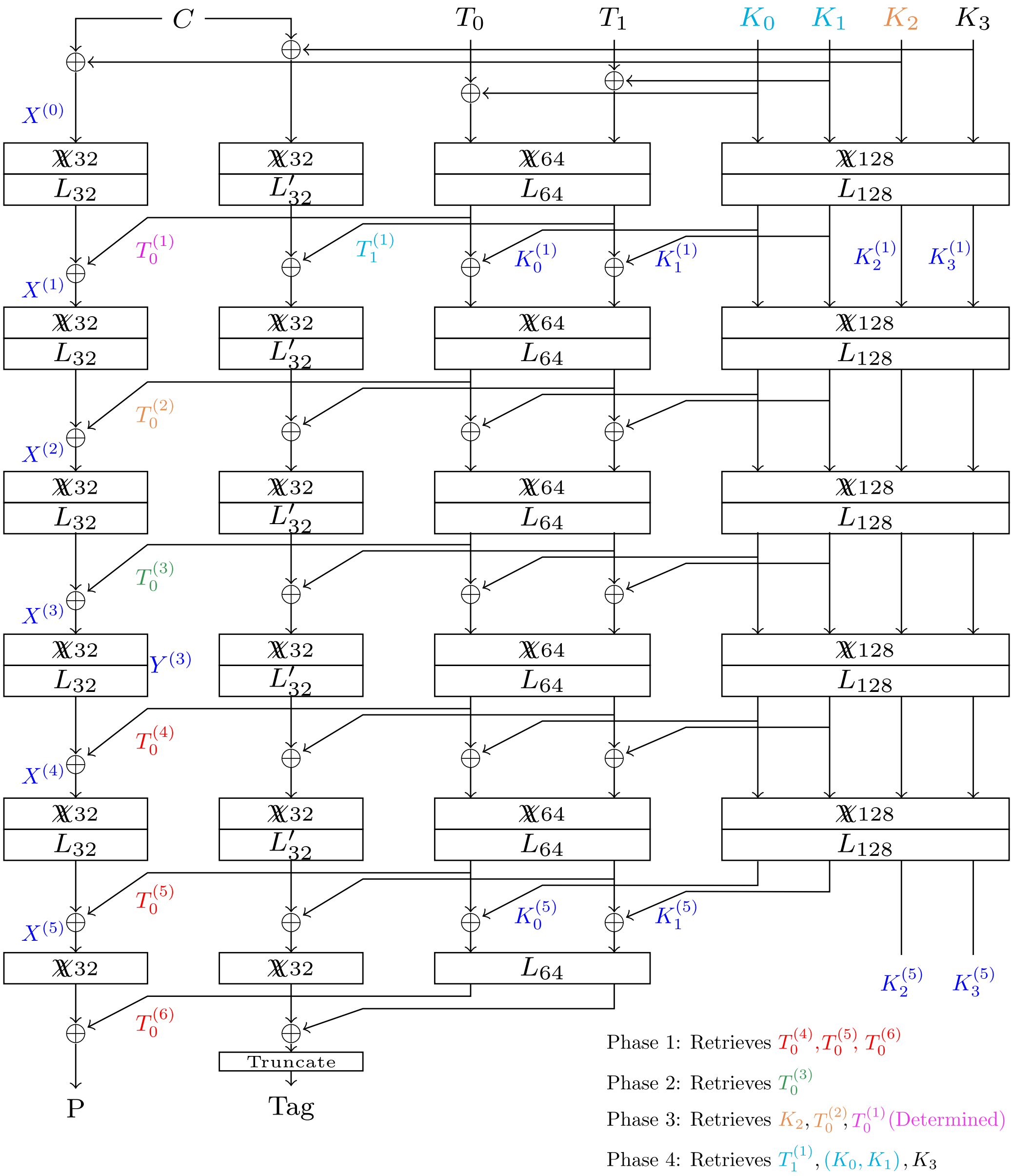

Figure 3: Attack on **6** rounds of `ChiLow`-$(32+\tau)$, and the core of our attack on 7 rounds.

Second phase: we aim to recover $T_0^{(3)}$ explointing a 2-round integral distinguisher that is extended four rounds forward. In this way, for the candidate values for $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}$ from the first phase, the parity will be checked for all 2^{32} possible values for $T_0^{(3)}$ and those leading to the balanced propoerty will be introduced as a candidate. The exploited distinguisher in this phase is:

$$\{21, 23\} \xrightarrow{\text{2-round}} \{0, ..., 31\}, \tag{17}$$

i.e., the input has 2 active bits, while the whole 32-bit output block is balanced. Hereafter, we refer to the distinguisher in Equation 17 as *Distinguisher III*.

It should be noted that the active bits in Distinguisher III is a subset of the active bits in Distinguisher I. So, a subset of the chosen ciphertexts in the first phase could be used for

Algorithm 1: Attack on 6-round ChiLow $(32+\tau)$: Phase 1 and Phase 2

Input: 3 distinct sets $[S_0, S_1, S_2]$ of 2^4 pairs (P, C) each, s.t. $C[0, ..., 20, 22, 24, 26, 28, ..., 31]$ takes a distinct constant in each set.
Output: Candidate tweak triples $\mathcal{T}'$.
$\mathcal{K} \leftarrow \emptyset$; $\mathcal{T} \leftarrow \emptyset$
Let $S_0^I \subset S_0$ be the 2^3 pairs of *Distinguisher I.*
Let $S_0^{III} \subset S_0$ be the 2^2 pairs of *Distinguisher III.*
for all $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}$ **do** // 1^{st} phase: prune tweak candidates
  PAR $\leftarrow 0$
  for all $(P, C) \in S_0^I$
    $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; PAR $\leftarrow$ PAR $\oplus X^{(3)}$
  endfor
  if PAR[2,3,14,25,26] $\neq 0$ **then continue with the next** $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}$
  for all $(P, C) \in S_0 \setminus S_0^I$
    $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; PAR $\leftarrow$ PAR $\oplus X^{(3)}$
  endfor
  if PAR $\neq 0$ **then continue with the next** $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}$
  for all $i \in \{1, 2\}$ **do**
    PAR$_i \leftarrow 0$
    for all $(P, C) \in S_i$
      $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; PAR$_i \leftarrow$ PAR$_i \oplus X^{(3)}$
    endfor
    if PAR$_i \neq 0$ **then continue with the next** $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}$
  endfor
  $\mathcal{T} \leftarrow \mathcal{T} \cup \{T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)}\}$
endfor
$\mathcal{T}' \leftarrow \emptyset$
for all $T_0^{(3)}$ **do** // 2^{nd} phase: extend to one more round
  for all $T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)} \in \mathcal{T}$ **do**
    PAR $\leftarrow 0$
    for all $(P, C) \in S_0^{III}$
      $X^{(2)} \leftarrow \mathcal{E}_4(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}, T_0^{(3)})$; PAR $\leftarrow$ PAR $\oplus X^{(2)}$
    endfor
    if PAR $= 0$ **then** $\mathcal{T}' \leftarrow \mathcal{T}' \cup \{T_0^{(6)} \| T_0^{(5)} \| T_0^{(4)} \| T_0^{(3)}\}$
  endfor
endfor
return $\mathcal{T}'$

the second phase, and no extra data is needed. We expect one $(= 1 \times 2^{32} \times 2^{-32})$ candidate on average for $T_0^{(6)} \,\|\, T_0^{(5)} \,\|\, T_0^{(4)} \,\|\, T_0^{(3)}$, due to the 32-bit filter and the 32-bit guess for $T_0^{(3)}$ over the candidates for $T_0^{(6)} \,\|\, T_0^{(5)} \,\|\, T_0^{(4)}$ from the first phase. Time complexity of this phase is estimated as $2^{33.5} (= 2^{32} \times 2^2 \times \frac{4}{6})$ encryptions.

Third phase: we guess $T_0^{(2)} \,\|\, K_2$ and compute $X^{(1)}$ by partial encryption under the candidates for $T_0^{(6)} \,\|\, T_0^{(5)} \,\|\, T_0^{(4)} \,\|\, T_0^{(3)}$ from the previous phase, and obtain $T_0^{(1)} = X^{(1)} \oplus L_{32}(\chi_{32}(C_1 \oplus K_2))$ for an arbitrary pair of (P, C). Then, we check the resulting candidates for $T_0^{(i)}, i = 1, ..., 6$ and K_2 using two other arbitrary pairs of (P, C) by $P = \mathcal{D}(C, T_0^{(6)}, ..., T_0^{(1)}, K_2)$. Hence, by $2^{65.6} (= 2^{64} \times 3)$ encryptions, we expect one candidate for $T_0^{(i)}, i = 1, ..., 6$ and K_2 on average.

Fourth phase: note that $T_0^{(1)} \,\|\, T_1^{(1)}$ depends only on the tweak T and 64 bits of the master key $K_0 \,\|\, K_1$. Hence, we guess the value for the 64-bit $K_0 \,\|\, K_1$ and filter out the guesses by the 32-bit condition imposed by $T_0^{(1)}$, recovered in the third phase. Afterward, we guess K_3 for each retrieved value for $K_0 \,\|\, K_1 \,\|\, K_2$ and filter them by the values recovered for $T_0^{(i)}$ for $i \in \{2, 3, 4, 5\}$ from the previous phases. The time complexity

Algorithm 2: Attack on 6-round `ChiLow` $(32{+}\tau)$: Phase 3 and Phase 4

35 **Input:** Candidate tweak triples $\mathcal{T}'$.
36 **Output:** Final key candidates $\mathcal{K}$.
37 $\mathcal{T} \leftarrow \emptyset$
38 Choose 3 distinct pairs $(P_i, C_i), i = 1, 2, 3.$
39 **for all** $T_0^{(2)} \parallel K_2$ **do** `// 3^{rd} phase: partial key recovery`
40   **for all** $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)} \parallel T_0^{(3)} \in \mathcal{T}'$ **do**
41         $X^{(1)} \leftarrow \mathcal{E}_4(P_1, T_0^{(6)}, T_0^{(5)}, T_0^{(4)}, T_0^{(3)}, T_0^{(2)})$
42         $T_0^{(1)} = L_{32}(\reflectbox{χ}_{32}(C_1 \oplus K_2)) \oplus X^{(1)}$
43   **if** $P_2 = \mathcal{D}(C_2, T_0^{(6)}, ..., T_0^{(1)}, K_2)$ *and* $P_3 = \mathcal{D}(C_3, T_0^{(6)}, ..., T_0^{(1)}, K_2)$ **then**
44             $\mathcal{T} \leftarrow \mathcal{T} \cup \{T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)} \parallel T_0^{(3)} \parallel T_0^{(2)} \parallel T_0^{(1)} \parallel K_2\}$

45 **for all** $(K_0 \parallel K_1)$ **do** `// 4^{th} phase: final key recovery`
46     $(\tilde{T}_0^{(1)} \parallel \tilde{T}_1^{(1)}) = L_{64}\Big(\reflectbox{χ}_{64}\big((K_0 \parallel K_1) \oplus (T_0 \parallel T_1)\big)\Big)$
47   **for all** $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)} \parallel T_0^{(3)} \parallel T_0^{(2)} \parallel T_0^{(1)} \parallel K_2 \in \mathcal{T}$ **do**
48   **if** $\tilde{T}_0^{(1)} \neq T_0^{(1)}$ **then continue with the next** $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)} \parallel T_0^{(3)} \parallel T_0^{(2)} \parallel T_0^{(1)} \parallel K_2$
49   **for all** K_3 **do**
50             $(K_0^{(0)} \parallel K_1^{(0)} \parallel K_2^{(0)} \parallel K_3^{(0)}) \leftarrow (K_0 \parallel K_1 \parallel K_2 \parallel K_3)$
51   **for** $i = 1, \ldots, 4$ **do**
52                 $(K_0^{(i)} \parallel K_1^{(i)} \parallel K_2^{(i)} \parallel K_3^{(i)}) =$
                  $L_{128}\Big(\reflectbox{$\chi$}_{128}\big((K_0^{(i-1)} \parallel K_1^{(i-1)} \parallel K_2^{(i-1)} \parallel K_3^{(i-1)}) \oplus c^{(i-1)}\big)\Big)$
53                 $(\tilde{T}_0^{(i+1)} \parallel \tilde{T}_1^{(i+1)}) = L_{64}\Big(\reflectbox{χ}_{64}\big((\tilde{T}_0^{(i)} \parallel \tilde{T}_1^{(i)}) \oplus (K_0^{(i)} \parallel K_1^{(i)})\big)\Big)$
54   **if** $\tilde{T}_0^{(i+1)} \neq T_0^{(i+1)}$ **then continue with the next** K_3
55             $\mathcal{K} \leftarrow \mathcal{K} \cup K$

56 **return** $\mathcal{K}$

of this phase is estimated as $2^{61.5} (= 2^{64} \times \frac{1}{6})$ decryptions for line 46 in algorithm 2 and $2^{64} (= 2^{64} \times 2^{-32} \times 2^{32})$ decryptions for line 52 and line 53 in algorithm 2. At the end of this phase, we expect to recover the correct master key, since we guess 3×32 bits $K_1 \parallel K_2 \parallel K_3$ in total and filter them by 5×32 bits $T_0^{(i)}, i \in \{1, 2, 3, 4, 5\}$ (line 48 and line 54 in algorithm 2).

Table 4: Summary of the key recovery attack on 6-round `ChiLow`-$(32{+}\tau)$

Phase	recovered parameter	No. of candidates	Time complexity
1	$T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{(4)}$	1	$2^{98.1} (\approx 2^{98} + 2^{93} + 2^{62})$ enc
2	$T_0^{(3)}$	1	$2^{33.5}$ enc
3	$T_0^{(2)} \parallel T_0^{(1)} \parallel K_2$	1	$2^{65.6}$ enc
4	$K_0 \parallel K_1 \parallel K_3$	1	$2^{64.23}$ dec

4.1.1 Complexity Analysis

The attack requires 3×2^4 chosen ciphertexts corresponding to three distinct sets of chosen ciphertexts with 4 active bits according to Distinguisher II, and the overall data complexity is $2^{5.6}$ chosen ciphertexts. As shown in Table 4, the time complexity is dominated by the first phase, which is estimated as $2^{98.1} (\approx 2^{98} + 2^{93} + 2^{62})$ encryptions. Only a negligible amount of memory is needed for temporary storage and intermediate computations

Algorithm 3: Improved Attack on 6-round `ChiLow` (32+τ): Phase 1 and Phase 2

Input: 5 distinct sets $[S_0, S_1, S_2, S_3, S_4]$ of 2^4 pairs (P, C) each, s.t. $C[0, ..., 20, 22, 24, 26, 28, ..., 31]$ takes a distinct constant in each set.
Output: Candidate tweak triples $\mathcal{T}'$.
$\mathcal{K} \leftarrow \emptyset$; $\mathcal{T} \leftarrow \emptyset$
Let $S_i^I \subset S_i$ be the 2^3 pairs of *Distinguisher I*; $i = 0, \ldots, 4$.
$\mathcal{S} \leftarrow \{S_i^I, S_i \setminus S_i^I | i = 0, \ldots, 4\}$; $T_0'^{(4)} \leftarrow 0$;
for all $T_0^{(6)} \| T_0^{(5)} \| T_0'^{(4)}[0, 2, 4, \ldots, 12, 17, 18, \ldots, 31]$ **do** // 2^{64+22}
  for all $S \in \mathcal{S}$ **do**
    `PAR` $\leftarrow 0$
    for all $(P, C) \in S$ // 2^3
      $Y^{(3)} \leftarrow L_{32}^{-1}(\mathcal{E}_2(P, T_0^{(6)}, T_0^{(5)})) \oplus T_0'^{(4)}$;
      if $Y^{(3)}[17, 19, 21, 23, 25, 27, 29, 31] = 0^8$ **then** check the next set in $\mathcal{S}$ // Invalid
      $X^{(3)} \leftarrow \chi^{-1}(Y^{(3)})$; `PAR` $\leftarrow$ `PAR` $\oplus X^{(3)}$
    endfor
    if `PAR[14]` $\neq 0$ **then** continue with the next $T_0^{(6)} \| T_0^{(5)} \| T_0'^{(4)}[\cdots]$
  endfor
  // End of step 1
  for all $T_0'^{(4)}[1, 3, 5, 7, 9, 11, 14, 16]$ **do** // 2^8
    for all $S \in \{S_0, S_1, S_2, S_3, S_4\}$ **do**
      `PAR` $\leftarrow 0$
      for all $(P, C) \in S$ // 2^4
        $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; `PAR` $\leftarrow$ `PAR` $\oplus X^{(3)}$
      endfor
      if `PAR`$[16] \neq 0$ **then** continue with the next $T_0'^{(4)}[14]$
    endfor
    // End of step 2
    for all $T_0'^{(4)}[13]$ **do**
      for all $S \in \{S_0, S_1, S_2, S_3, S_4\}$ **do**
        `PAR` $\leftarrow 0$
        for all $(P, C) \in S$ // 2^4
          $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; `PAR` $\leftarrow$ `PAR` $\oplus X^{(3)}$
        endfor
        if `PAR`$[16] \oplus$ `PAR`$[13] \neq 0$ **then** continue with the next $T_0'^{(4)}[13]$
      endfor
      // End if step 3
      for all $T_0'^{(4)}[15]$ **do**
        for all $S \in \{S_0, S_1, S_2, S_3, S_4\}$ **do**
          `PAR` $\leftarrow 0$
          for all $(P, C) \in S$ // 2^4
            $X^{(3)} \leftarrow \mathcal{E}_3(P, T_0^{(6)}, T_0^{(5)}, T_0^{(4)})$; `PAR` $\leftarrow$ `PAR` $\oplus X^{(3)}$
          endfor
          if `PAR` $\neq 0$ **then** continue with the next $T_0'^{(4)}[15]$
        endfor
      endfor
    endfor
    $\mathcal{T} \leftarrow \mathcal{T} \cup \{T_0^{(6)} \| T_0^{(5)} \| T_0'^{(4)}\}$
  endfor
endfor
return $\mathcal{T}'$

4.2 Improved 6-round Attack

In this section, we improve the proposed attack in Subsection 4.1 by making some savings in tweak guesses during the first phase, which is the dominant phase of the attack. Rather than guessing all tweaks simultaneously, we guess them incrementally, adding subsequent guesses only after filtering the candidates.

The first phase of the previous attack will proceed in four steps as described in this section and summarized in Algorithm 3. We define $T_0^{\prime(4)} = L_{32}^{-1}(T_0^{(4)})$ as the equivalent tweak of round 4. Let S_0, S_1, S_2, S_3, and S_4 be the five data sets for *Distinguisher II*. From these, we derive 10 data sets for *Distinguisher I* by fixing the ciphertext bit $C[27]$ to a constant value in each set. In more detail, we define $S_i^I = S_i|_{Ctx[27]=0}$, and $S_i \setminus S_i^I = S_i|_{Ctx[27]=1}$, each of which provides an appropriate set for *Distinguisher I*. Let $\mathcal{S} = \{S_i^I, S_i \setminus S_i^I | i = 0, \dots, 4\}$ denote the set of all 10 sets for *Distinguisher I*.

The description of $x = \chi_{32}^{-1}(y)$ for $i = 13, 14, 15, 16$ is provided in Appendix A. According to Equation 29, x_{14} depends linearly on y_{16}. Furthermore, it is totally independent of $y_{14}, y_1, y_3, \dots, y_{13}, y_{15}$ under the condition below:

$$\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27}\overline{y}_{29}\overline{y}_{31} = 0 \tag{18}$$

which holds with a probability of $p = 1 - 2^{-8}$ assuming that the variables involved in Equation 18 are uniformly random.

Step 1. We guess the values of $T_0^{(6)} \parallel T_0^{(5)}$ together with the 22 bits of $T_0^{\prime(4)}$ that significantly affect $X^{(3)}[14]$, namely $T_0^{\prime(4)}[0, 2, 4, \dots, 12, 17, 18, \dots, 31]$. For each data set corresponding to *Distinguisher I* in $\mathcal{S}$, we first verify that the independence condition (Equation 18) holds for all 2^3 pairs within the set. If the condition is satisfied, we proceed to compute $\texttt{PAR}[14]$; otherwise, we skip to the next data set. If $\texttt{PAR}[14] \neq 0$, the current guess for the round tweak is incorrect, and we move on to the next candidate for the guessed tweaks.

For each set, we process pairs and stop at the first "invalid" pair, that is defined as $Y^{(3)}[17, 19, 21, 23, 25, 27, 29, 31] = 0^8$. This stop event occurs with probability 2^{-8} per pair. Hence, with $p = (1 - 2^{-8})$ the expected number of executions of line 10 *per set* is:

$$\mathbb{E}\big[\#\text{line 10 per set}\big] = \sum_{j=0}^{7} p^j = \frac{1 - p^8}{1 - p} = 2^8\big[1 - (1 - 2^{-8})^8\big] = 2^{2.98}. \tag{19}$$

However, because of the condition in line 14 (which holds with probability 2^{-1}), the probability that a set S ends the search for the current tweak is:

$$r \;=\; \Pr(\text{set is valid}) \cdot \Pr\big(\texttt{PAR}[14] \neq 0 \,\big|\, \text{set is valid}\big) = (1 - 2^{-8})^8 \cdot 2^{-1} = 2^{-1.05}. \tag{20}$$

Thus, the number of sets processed per tweak is:

$$\mathbb{E}[\#\text{sets per tweak}] = \sum_{j=0}^{9} (1 - r)^j = \frac{1 - (1 - r)^{10}}{r} = 2^{1.04}. \tag{21}$$

As a result, the total time complexity for executing line 10, in terms of the encryption operation is:

$$\begin{aligned} T_0 \;&=\; 2^{86} \times \mathbb{E}\big[\#\text{line 10 per set}\big] \times \mathbb{E}[\#\text{sets per tweak}] \times \frac{1}{3} \\ &=\; 2^{86} \times 2^{2.98} \times 2^{1.04} \times \frac{1}{3} \approx 2^{88.44} \text{ enc.} \end{aligned} \tag{22}$$

where the factor $\frac{1}{3}$ is used to normalize the computation of the 2-round encryption in line 10 to the 6-round encryption. Moreover, within a set, line 12 is executed once per valid pair before the first invalid pair appears, with a maximum of 8 pairs, i.e.

$\mathbb{E}\big[\#\text{line 12 per set}\big] = \sum_{j=1}^{8} p^j = 2^{2.97}$. As a result, the total time spent executing line 12 is:

$$\begin{aligned} T_1 &= 2^{86} \times \mathbb{E}\big[\#\text{line 12 per set}\big] \times \mathbb{E}\big[\#\text{sets per tweak}\big] \times \frac{1}{6} \\ &= 2^{86} \times 2^{2.97} \times 2^{1.04} \times \frac{1}{6} \approx 2^{87.43} \text{ enc.} \end{aligned} \quad (23)$$

where $\frac{1}{6}$ normalizes the computation of a single χ_{32}^{-1} in line 12 to the 6-round encryption.

Now, we compute the expected number of tweak candidates delivered to the next steps of algorithm 3. We showed that a single set rejects a wrong tweak with probability r (see Equation 20). Therefore, a wrong value for $T_0^{(6)} \| T_0^{(5)} \| T_0^{\prime(4)}[0, 2, 4, \ldots, 12, 17, 18, \ldots, 31]$ survives a set with probability $1 - r$, and 10 sets with probability $(1 - r)^{10}$. Hence, the number of tweaks delivered to the next steps is:

$$N_1 = 2^{86} \left(1 - r\right)^{10} \approx 2^{76.44}. \quad (24)$$

Step 2. For each candidate value of $T_0^{(6)} \parallel T_0^{(5)} \parallel T_0^{\prime(4)}[0, 2, 4, \ldots, 12, 17, 18, \ldots, 31]$ that survives from Step 1, we further guess 8 bits of $T_0^{\prime(4)}[1, 3, 5, 7, 9, 11, 14, 16]$. For each of the five sets of *Distinguisher II*, we then compute $\texttt{PAR}[16]$. If $\texttt{PAR}[16] \neq 0$, the guessed tweak is discarded and the process continues with the next candidate. According to (Equation 29), x_{16} is independent of y_{15} and depends linearly on y_{13}. Consequently, it is unnecessary to guess $T_0^{\prime(4)}[13, 15]$ when computing the parity of the XOR sum over $X^{(3)}[16]$. Moreover, we emphasize that the computation involves only 3 rounds of encryption (out of the 6 total rounds). Therefore, a factor of $\frac{1}{2}$ is applied when evaluating the time complexity. The resulting time complexity and the number of surviving candidates at the end of this step are as follows:

$$\begin{aligned} T_2 &= \frac{1}{2} \times 2^{76.44} \times 2^8 \times 2^4 \times [1 + 2^{-1} + 2^{-2} + 2^{-3} + 2^{-4}] = 2^{88.39} \text{ enc} \\ N_2 &= 2^{76.44} \times 2^8 \times 2^{-5} = 2^{79.44} \end{aligned} \quad (25)$$

Step 3. According to (29), $x_{16} \oplus x_{13}$ linearly depends on y_{15}. So, we need only to guess $T_0^{\prime(4)}[13]$ to compute and check the balance of $\texttt{PAR}[16] \oplus \texttt{PAR}[13]$. The time complexity and the number of candidates remaining at the end of this step are as follows.

$$\begin{aligned} T_3 &= \frac{1}{2} \times 2^{79.44} \times 2 \times 2^4 \times [1 + 2^{-1} + 2^{-2} + 2^{-3} + 2^{-4}] = 2^{84.39} \text{ enc} \\ N_3 &= 2^{79.44} \times 2 \times 2^{-5} = 2^{75.44} \end{aligned} \quad (26)$$

Step 4. We guess the only remaining tweak bit, $T_0^{\prime(4)}[15]$, and compute the value of $\texttt{PAR}$ for all coordinates of $X^{(3)}$ except for $14, 16, 13$, each of which provides a one-bit filter. So, at the end of this phase, the time complexity and the remaining candidates would be as follows.

$$\begin{aligned} T_4 &= \frac{1}{2} \times 2^{75.44} \times 2 \times 2^4 \times \sum_{j=0}^{4} 2^{-29 \times j} = 2^{79.44} \text{ enc} \\ N_4 &= 2^{75.44} \times 2 \times 2^{-4 \times 29} \ll 1 \text{ (Only the correct round tweaks remain)} \end{aligned} \quad (27)$$

Therefore, the total time complexity of the first phase (algorithm 3) is:

$$T_{ph.1} = T_0 + T_1 + T_2 + T_3 + T_4 = 2^{88.44} + 2^{87.43} + 2^{88.39} + 2^{84.39} + 2^{79.44} \approx 2^{89.78} \text{ enc.} \quad (28)$$

The attack requires $5 \times 2^4 = 2^{6.32}$ chosen ciphertexts with an improvement of approximately 8.32 bits over the primary 6-round attack, and can be straightforwardly extended to 7 rounds.

4.3 7-round Attack

The improved 6-round attack presented in Subsection 4.2 can be directly extended to 7 rounds of `ChiLow`-(32+τ). For the 7-round attack, in addition to guessing $T_0^{(6)} \| T_0^{(5)} \| T_0'^{(4)}[0, 2, 4, \ldots, 12, 17, 18, \ldots, 31]$, we also guess the 32 bits of $T_0^{(7)}$ used in line 6 of algorithm 3. Consequently, the time complexity and the number of filtered tweak candidates at each step of the first phase increase by a factor of 2^{32}. Thus, the total time for the first phase of the attack, $T_{ph.1}$ given in Equation 28, is multiplied by 2^{32} as well, which would be $T_{ph.1} = 2^{121.78}$

We note, however, that despite this increase, we still expect only the correct round tweaks $T_0^{(7)} \| T_0^{(6)} \| T_0^{(5)} \| T_0'^{(4)}$ to reach the second phase. That is, N_4 in Equation 27 remains below one, so the time complexities of subsequent phases are not significantly affected. As before, the overall time complexity remains dominated by the first phase, and the data requirement is unchanged. Therefore, the 7-round attack requires $2^{6.32}$ chosen ciphertexts and has a time complexity of $2^{121.78}$.

5 Conclusion

We investigated the security of `ChiLow`-(32+τ), an instantiation of the ACE-2 construction, against integral attacks. We present several integral distinguishers for 2, 3, and 3.5 rounds, along with key-recovery attacks on 6 and 7 out of the 8 rounds. All proposed distinguishers and key-recovery attacks adhere to the strict query limits specified by the designers. Both the 6-round and 7-round attacks require $2^{6.32}$ chosen ciphertexts with negligible memory. The time complexities of the 6-round and 7-round attacks are $2^{89.78}$ and $2^{121.78}$, respectively. Furthermore, we successfully addressed the challenge introduced by the designers concerning the recovery of the master key from the later-round recovered tweaks.

References

[ABD+24] Parisa Amiri-Eliasi, Yanis Belkheyar, Joan Daemen, Santosh Ghosh, Daniël Kuijsters, Alireza Mehrdad, Silvia Mella, Shahram Rasoolzadeh, and Gilles Van Assche. Koala: A low-latency pseudorandom function. In Maria Eichlseder and Sébastien Gambs, editors, *SAC 2024*, volume 15517 of *LNCS*, pages 239–266. Springer, 2024. doi:10.1007/978-3-031-82841-6_10.

[BDG+25] Yanis Belkheyar, Patrick Derbez, Shibam Ghosh, Gregor Leander, Silvia Mella, Léo Perrin, Shahram Rasoolzadeh, Lukas Stennes, Siwei Sun, Gilles Van Assche, et al. Chilow and chichi: new constructions for code encryption. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 212–243. Springer, 2025.

[BDPA13] Guido Bertoni, Joan Daemen, Michaël Peeters, and Gilles Van Assche. Keccak. In Thomas Johansson and Phong Q. Nguyen, editors, *EUROCRYPT 2013,*, volume 7881 of *LNCS*, pages 313–314. Springer, 2013. doi:10.1007/978-3-642-38348-9_19.

[Dae95] Joan Daemen. *Cipher and hash function design strategies based on linear and differential cryptanalysis.* PhD thesis, Doctoral Dissertation, March 1995, KU Leuven, 1995.

[DEMS21] Christoph Dobraunig, Maria Eichlseder, Florian Mendel, and Martin Schläffer. Ascon v1.2: Lightweight authenticated encryption and hashing. *J. Cryptol.*, 34(3):33, 2021. doi:10.1007/S00145-021-09398-9.

[DKR97] Joan Daemen, Lars R. Knudsen, and Vincent Rijmen. The block cipher square. In Eli Biham, editor, *FSE '97,*, volume 1267 of *LNCS*, pages 149–165. Springer, 1997. doi:10.1007/BFB0052343.

[GWZ25] Jian Guo, Shichang Wang, and Tianyu Zhang. Breaking full ChiLow-32. Cryptology ePrint Archive, Paper 2025/1648, 2025. URL: https://eprint.iacr.org/2025/1648.

[KW02] Lars R. Knudsen and David A. Wagner. Integral cryptanalysis. In Joan Daemen and Vincent Rijmen, editors, *FSE 2002,*, volume 2365 of *LNCS*, pages 112–127. Springer, 2002. doi:10.1007/3-540-45661-9_9.

[Lai94] Xuejia Lai. Higher order derivatives and differential cryptanalysis. In *Communications and cryptography*, pages 227–233. Springer, 1994.

[LNV25] Eran Lambooij, Patrick Neumann, and Michiel Verbauwhede. Meet-in-the-middle attacks on full ChiLow-32. Cryptology ePrint Archive, Paper 2025/1601, 2025. URL: https://eprint.iacr.org/2025/1601.

[PHH+25] Shuo Peng, Jiahui He, Kai Hu, Zhongfeng Niu, Shahram Rasoolzadeh, and Meiqin Wang. Cryptanalysis of ChiLow with cube-like attacks. Cryptology ePrint Archive, Paper 2025/1581, 2025. URL: https://eprint.iacr.org/2025/1581.

[TM16] Yosuke Todo and Masakatu Morii. Bit-based division property and application to simon family. In *International Conference on Fast Software Encryption*, pages 357–377. Springer, 2016.

[Tod15] Yosuke Todo. Structural evaluation by generalized integral property. In *Annual International Conference on the Theory and Applications of Cryptographic Techniques*, pages 287–314. Springer, 2015.

[XZBL16] Zejun Xiang, Wentao Zhang, Zhenzhen Bao, and Dongdai Lin. Applying milp method to searching integral distinguishers based on division property for 6 lightweight block ciphers. In *International conference on the theory and application of cryptology and information security*, pages 648–678. Springer, 2016.

A Description of χ_{32}^{-1} at the middle coordinates

Based on Lemmas 3 to 6 of [BDG+25], the description of $x = \chi_{32}^{-1}(y)$ for coordinates $i = 13, 14, 15, 16$ is derived as follows (x_i denotes the i^{th} bit of x, with LSB x_0 at the rightmost).

$$
\begin{aligned}
x_{16} = {} & y_{13} + \big(y_0 + y_2\overline{y}_1 + y_4\overline{y}_1\overline{y}_3 + y_6\overline{y}_1\overline{y}_3\overline{y}_5 + y_8\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7 \\
& + y_{10}\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7\overline{y}_9 + y_{12}\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7\overline{y}_9\overline{y}_{11}\big) \\
& \cdot \Big(\overline{y}_{16} + \big(y_{18}\overline{y}_{17} + y_{20}\overline{y}_{17}\overline{y}_{19} + y_{22}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21} \\
& \quad + y_{24}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23} + y_{26}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25} \\
& \quad + y_{28}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27} + y_{30}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27}\overline{y}_{29}\big) \\
& \quad + y_{14}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27}\overline{y}_{29}\overline{y}_{31}\Big), \\
x_{16} + x_{13} = {} & y_{15} + \Big(\overline{y}_{13} + \overline{y}_{16}(y_0 + y_2\overline{y}_1 + y_4\overline{y}_1\overline{y}_3 + y_6\overline{y}_1\overline{y}_3\overline{y}_5 \\
& \quad + y_8\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7 + y_{10}\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7\overline{y}_9 + y_{12}\overline{y}_1\overline{y}_3\overline{y}_5\overline{y}_7\overline{y}_9\overline{y}_{11})\Big) \\
& \cdot \big(y_{17} + y_{19}\overline{y}_{18} + y_{21}\overline{y}_{18}\overline{y}_{20} + y_{23}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22} \\
& \quad + y_{25}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24} + y_{27}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26} \\
& \quad + y_{29}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26}\overline{y}_{28} + y_{31}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26}\overline{y}_{28}\overline{y}_{30}\big), \\
x_{15} = {} & y_{14} + \big(y_1\overline{y}_0 + y_3\overline{y}_0\overline{y}_2 + y_5\overline{y}_0\overline{y}_2\overline{y}_4 + y_7\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6 \\
& \quad + y_9\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8 + y_{11}\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8\overline{y}_{10}\big) \\
& + \big(\overline{y}_{15} + (\overline{y}_{17} + y_{19}\overline{y}_{18} + y_{21}\overline{y}_{18}\overline{y}_{20} + y_{23}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22} \\
& \quad + y_{25}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24} + y_{27}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26} \\
& \quad + y_{29}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26}\overline{y}_{28} + y_{31}\overline{y}_{18}\overline{y}_{20}\overline{y}_{22}\overline{y}_{24}\overline{y}_{26}\overline{y}_{28}\overline{y}_{30})\overline{y}_{13}\big) \\
& \cdot (\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8\overline{y}_{10}\overline{y}_{12}), \\
x_{14} = {} & y_{16} + \big(y_{18}\overline{y}_{17} + y_{20}\overline{y}_{17}\overline{y}_{19} + y_{22}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21} \\
& \quad + y_{24}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23} + y_{26}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25} \\
& \quad + y_{28}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27} + y_{30}\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27}\overline{y}_{29}\big) \\
& + \big(y_{14} + (y_1\overline{y}_0 + y_3\overline{y}_0\overline{y}_2 + y_5\overline{y}_0\overline{y}_2\overline{y}_4 + y_7\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6 \\
& \quad + y_9\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8 + y_{11}\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8\overline{y}_{10} + y_{13}\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8\overline{y}_{10}\overline{y}_{12}) \\
& \quad + y_{15}\overline{y}_0\overline{y}_2\overline{y}_4\overline{y}_6\overline{y}_8\overline{y}_{10}\overline{y}_{12}\big) \\
& \cdot (\overline{y}_{17}\overline{y}_{19}\overline{y}_{21}\overline{y}_{23}\overline{y}_{25}\overline{y}_{27}\overline{y}_{29}\overline{y}_{31}).
\end{aligned}
\tag{29}
$$

Equation 29 shows that x_{16} is independent of y_{15} and depends linearly on y_{13}, while a similar relationship holds for $x_{13} \oplus x_{16}$ with regrad to y_{14} and y_{15}, respectively. In the cases of x_{14} and x_{15}, there are no independent bits of y. However, besides the linear dependencies on y_{16} and y_{14}, respectively, there exist weak dependencies where certain y bits appear only once with a high degree (greater than 7 or 8). All these dependencies are summarized in Table 5.

Table 5: Dependency of x coordinates on y coordinates

x coordinate	Independence w.r.t.	Linear dependency	week dependency
x_{14}	-	y_{16}	$y_{14}, y_1, y_3, \ldots, y_{13}, y_{15}$ (degree ≥ 8)
x_{15}	-	y_{14}	$y_{15}, y_{17}, \ldots, y_{31}$ (degree ≥ 7)
x_{16}	y_{15}	y_{13}	-
$x_{16} + x_{13}$	y_{14}	y_{15}	-